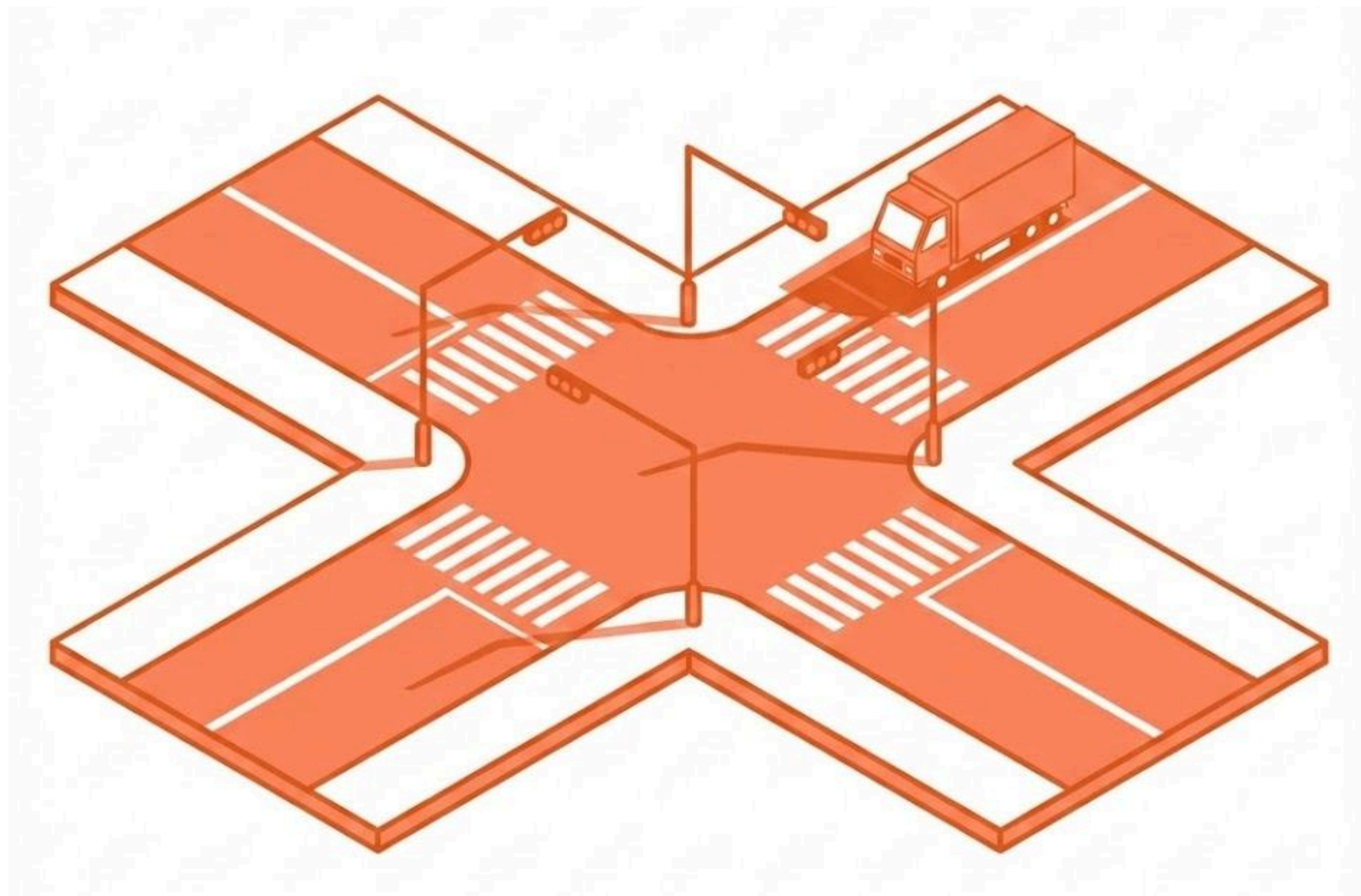


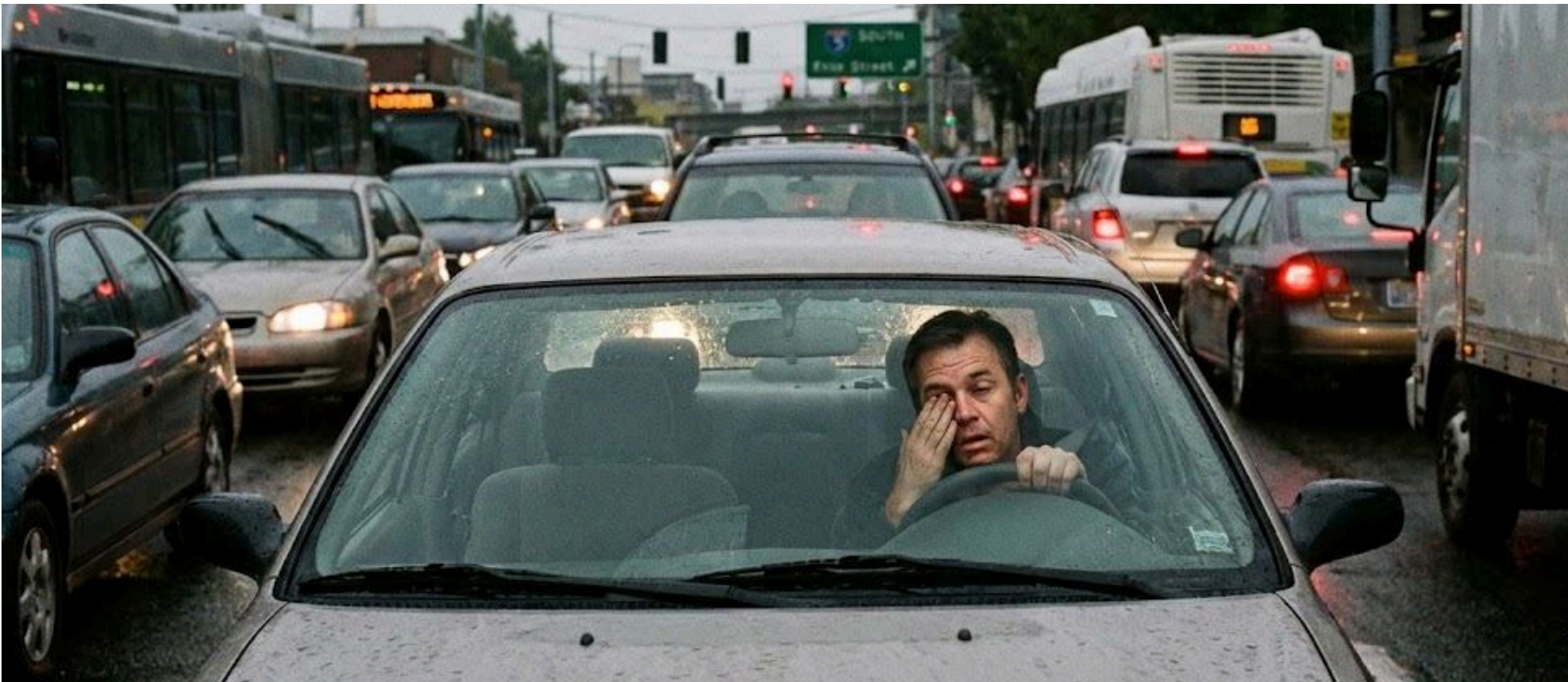
Smart Local Traffic Light Management System

2025 Winter Global Education Program - Group 10



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Economic Impact of Traffic Congestion in Seattle Source



INRIX 2024 Global Traffic Scorecard

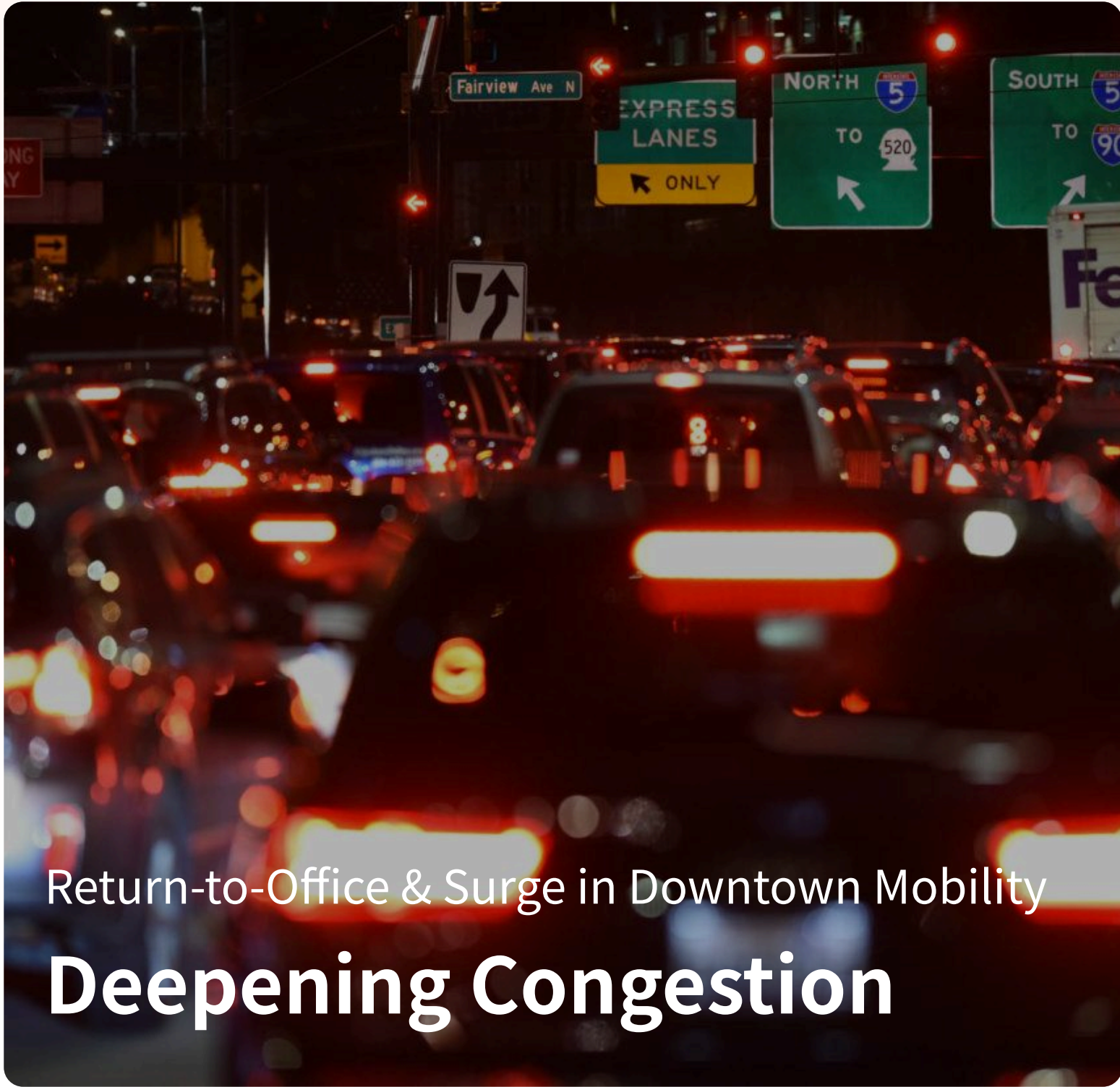
The Big Numbers

68 Hourse	\$1,252	\$2.0 Billion
Annual Hours Lost per Driver	Annual Opportunity Cost per Driver	Seattle Regional Economic Loss
(40% higher than the US average)	(Converted value of fuel & time)	(Based on late 2025 Axios estimates)

Traffic congestion in Seattle is rising again due to Return-to-Office (RTO) mandates and increased downtown mobility. As of 2024, reports indicate that drivers **lose an average of 63** hours annually, incurring approximately \$1,128 in costs per person.

For the city as a whole, the cost of congestion is estimated at \$1.8 billion. This congestion leads to **lost productivity** and **regional economic damage**. Furthermore, as pressure concentrates on bottleneck sections, **the predictability of daily travel significantly decreases**.

The Necessity of Intelligent Signal Systems



Return-to-Office & Surge in Downtown Mobility
Deepening Congestion

Recorded a 9% increase in congestion in 2024. Downtown traffic volume has reverted to pre-pandemic levels due to the reduction of hybrid work policies, significantly exacerbating bottleneck phenomena. (INRIX, 2024)

INRIX 2024 Global Traffic Scorecard / PR Newswire



Consistent Deterioration of Key Metrics
Cross-Verified Stagnation

Confirmed by both INRIX and the TomTom Index: Travel time per 10km has increased, with a 31% congestion level recorded. Severe bottlenecks in specific sections (e.g., I-405) undermine the travel reliability of the entire city. (TomTom, 2024)

Axios Seattle (2025-12-01): "Seattle drivers lose more time to traffic than national average"



Potential for Immediate Improvement via Signal Optimization
A Proven Solution

SDOT pilot results confirm an 11% improvement in travel time and up to 20% improvement in reliability upon introducing adaptive signal control. This proves that data-driven signal optimization is a lower-cost, higher-efficiency solution compared to simple infrastructure expansion. (SDOT, 2024)

TomTom Traffic Index 2024 (Seattle)

“Traffic congestion in Seattle is causing \$2 billion in annual economic losses.

We propose **a smart city strategy** that increases the predictability of movement and reduces carbon emissions." By expanding the proven "adaptive signal optimization" technology

| Personas, Goals, and Pain Points

A) Urban commuter driver
Bellevue–Seattle area



B) Traffic engineer / signal operations manager
City Hall / DOT

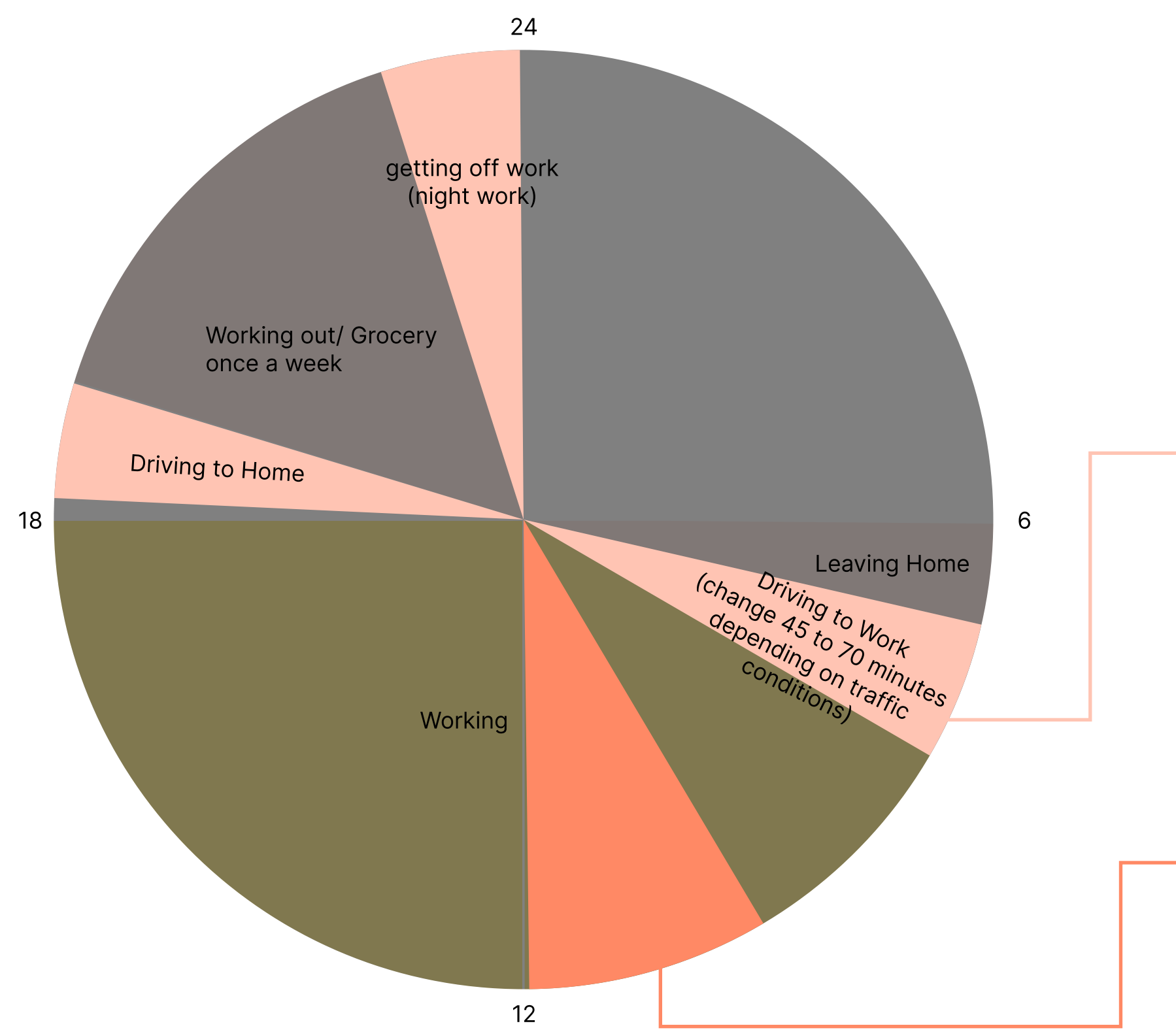


C) Pedestrian + public transit user
bus / King County connection



Informations	• Name: Jason Miller • Age/Gender: 45 / Male • Residence: Single-family home near Eastgate / Factoria • Occupation: Engineering Program Manager at Microsoft (Redmond Campus) • Income: \$150,000 / year • Family: Spouse + 1 middle school child • Transportation: Personal Vehicle (Subaru Outback)	• Name: Linda Hamilton • Age/Gender: 50 / Female • Residence: Homeowner in the Somerset neighborhood • Workplace: City of Bellevue Transportation Department • Job Title: Senior Traffic Operations Manager • Income: \$160,000 / year	• Name: Robert Frost • Age/Gender: 72 / Male • Residence: Condo near a senior community in Wilburton • Occupation: Retired (Former Retail Manager) • Income: Pension and savings (Lower-middle level) • Transportation: Walking and Bus
Goals	Predictable commute time, minimize unnecessary stops/waiting, “feel” a green wave.	Safety (accident prevention), congestion relief, fewer complaints, comply with budget/procurement constraints, audit-ready records (logs).	Safe and fast crossing, reliability so they don't miss bus transfers.
Pain Point	Long waits at major intersections, consecutive red lights (“red wave”), sudden spillover congestion, confusion/accident risk during power outages.	Compatibility with legacy controllers (e.g., NEMA TS2), deployment & maintenance burden, demand for explainability (why timing changes), handling events/construction, need for manual control when failures occur.	Inconvenient walk-signal request method / excessive waiting time, short crossing time, missing the button timing can cause several minutes of waiting, feeling neglected due to car-first policies.

• Traffic Usage Pattern Analysis



• Weekly Basis(“How Much/When”)

Frequency of Vehicle Use: 6 days/week, including 2~4 short trips daily
Total Weekly Driving Time: Approx. 9~12 hours/week
Total Weekly Driving Distance: Approx. 180~260 km/week

Most frequent Driving Time

Morning Commute: 07:20~08:10
Evening Commute: 18:10~19:30
Returning from Overtime: 22:00~23:30 (Core Pain Point Period)

Weekend Travel

- Saturday Morning: 10:00~12:00(Grocery/Cafe/Park)
- Sunday : Mostly stay near home, minimal driving

| Scenario Development via Persona Analysis

Situation
Description

Time
/Weather

Place and
situation

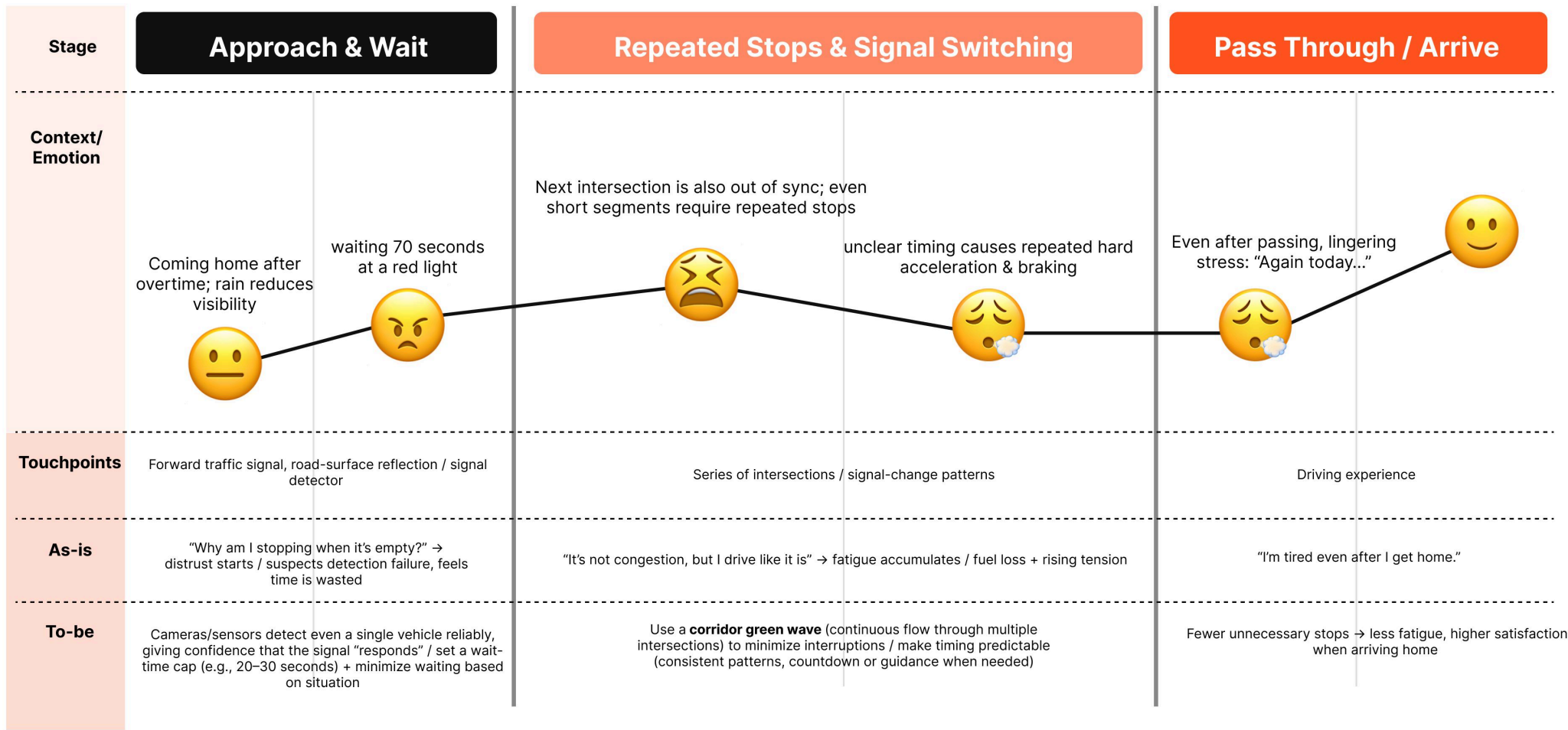
The core
of probelm

Driver’s point of view

Tuesdays 23:10, Rain, Empty street

a large intersection in Vallevue

Red light stays long like a fixed siganl even though there are few cars, and “broken” is repeated at each intersection, resulting in fatigue accumulation

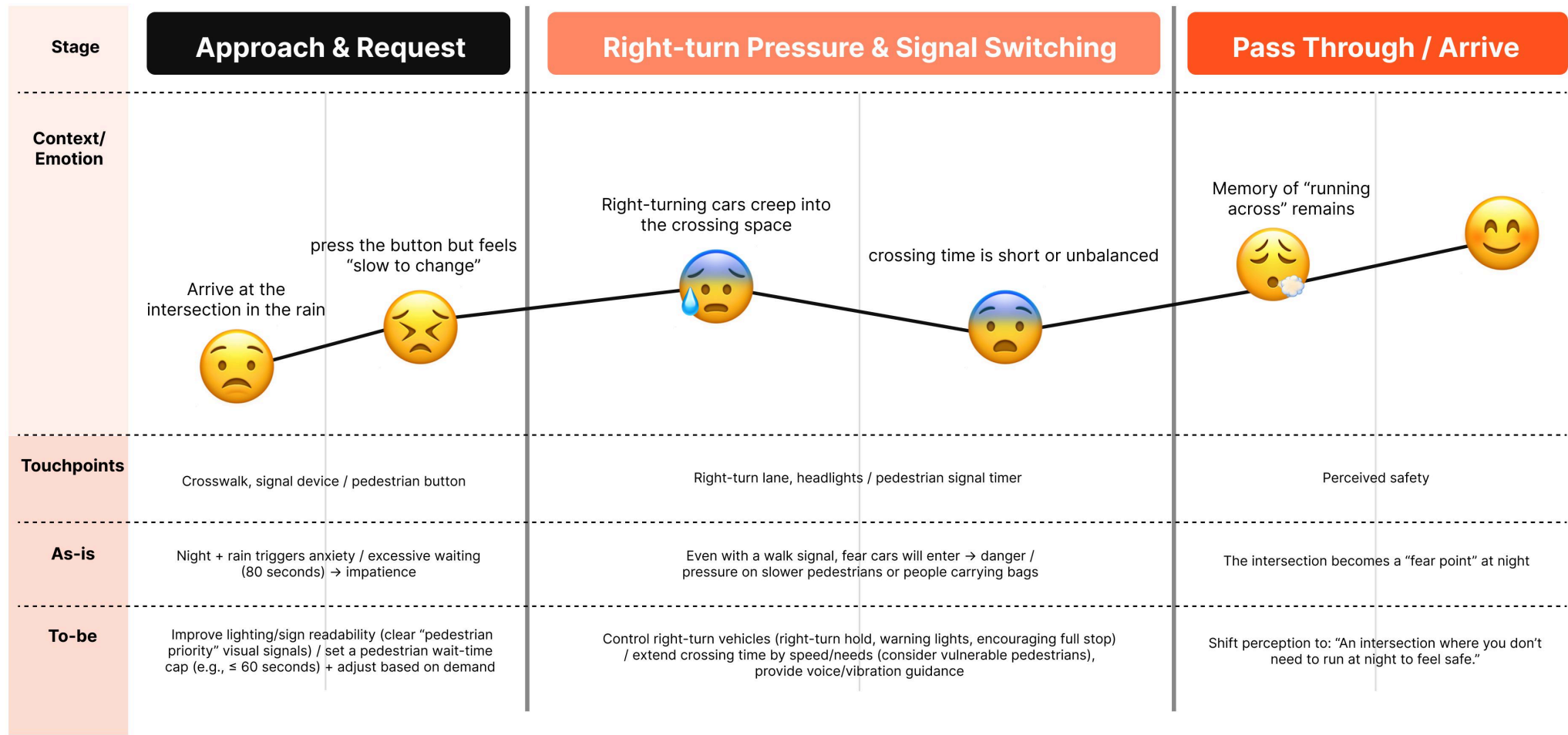


Pedestrian point of view

Thursday 20:50, Rainy Night

Return home after walking, 80 seconds at intersection, right turn car coming in front of me

Long wait times, psychological threat grow due to eye fight with right turn vehicle, “Not sure if car will stop” if signal goes out



As-is →
To-be Effect

Eliminating unnecessary red light atmosphere(Late Night, Low Demand Time Optimization)
Reduce rapid acceleration/quick bracking → Reduce fuel + reduce risk of accidents
Crossroads Offbeat Reduction(Continuous Flow) → Remove the feeling of “none-Stalled”

Result: “Late Night Alone Stoppable” disappears, resulting in a significant reduction in fatigue and irritation

Mitigating the problem of excessive walking wating times
Reduce right turn vehicle pressure (actually guarantee “authority” of the pedestrian signal)
Securing psychological safety at night and in case of rain

Hypothesis I Vehicle Delay Reduction

Hypothesis Definition

The signal control algorithm can monitor real-time traffic conditions (vehicle inflow and queue length) using computer vision (CV), and dynamically adjust the green-light duration in proportion to these conditions. This enables a significant reduction in the average vehicle delay per cycle compared to a fixed-period baseline.
(Target : 20% + improvement)

Metric Definition

- Average vehicle delay is defined as the arithmetic mean of the time difference between “actual continuous passage time(T_{actual})”
- and “free-flow passage time(T_{free})” across N measured segments.

$$D_{avg} = \frac{1}{N} \sum_{k=1}^N (T_{actual} - T_{free})$$

Validation Method

- Configure SUMO simulation environments that reflect peak commuting conditions (weekdays 8 AM, 12PM, etc).
- Run simulations repeatedly to compare the baseline signal system and the MVP signal prototype, and compute average vehicle delay for each case.
- Validate the hypothesis by checking whether the MVP system achieves a 20%+reduction in average vehicle delay while maintaining pedestrian SLA constraints.

Success Criteria

- Phase-1 Goal : 20% reduction in vehicle delay
- Constraint : Maintain pedestrian maximum wait time ≤ 90 seconds

Hypothesis II

Pedestrian Wait-Time Reduction

Hypothesis Definition

Instead of simply extending the duration of the pedestrian green signal in intersection control, if the maximum pedestrian wait time is limited to 90 seconds or less (SLA constraint), pedestrian requests are processed based on this SLA, and vehicle signal time is redistributed using weights according to real-time traffic conditions, then the average wait time for randomly arriving pedestrians will show a statistically significant reduction compared to the baseline.

- (Supplementary Hypothesis/Constraint): While satisfying the above conditions, the increase in average vehicle delay can be maintained within an acceptable range ($\leq 5\%$).

Metrics Definition

- **Primary Metric:** Pedestrian Average Wait Time
- Definition: Time elapsed from the moment a pedestrian crossing request is made until the pedestrian walk signal begins.
- Formula:

$$\bar{W}_{ped} = \frac{1}{N} \sum_{k=1}^N (t_k^{walk} - t_k^{request})$$

- **Secondary Metrics (Distribution)**
- Pedestrian Wait Time 95th Percentile (P95 Ped Wait)
 - Definition: The value at the top 95th percentile of the pedestrian wait time distribution.
 - Purpose: To prevent the phenomenon where "the average decreases, but some users wait excessively long."
- Maximum Pedestrian Wait Time (Max Ped Wait)
 - Definition: The longest wait time recorded during the measurement period.
 - Purpose: To confirm compliance with the SLA.
- **Vehicle Impact (Trade-off Verification)**
 - Vehicle Average Delay
 - Definition: Arithmetic mean of (Actual Travel Time) – (Free-Flow Travel Time).
 - Purpose: To ensure pedestrian optimization does not excessively degrade vehicle flow.
 - Note: Free-flow travel time is defined as the time taken to pass without stopping.

Validation Method

- Set up a SUMO simulation environment tailored to scenarios for peak (08:00) and off-peak (21:00) hours.
- Apply Baseline and H2 MVP signal controls, repeat the executions, and collect pedestrian/vehicle metric logs for each run.
- Verify the hypothesis based on the collected results.
 - If the average pedestrian wait time satisfies the maximum wait time constraint of 90 seconds and decreases by more than 15% compared to the baseline, the hypothesis is considered True.
- Statistical Test
 - Reason: The wait time distribution is expected to be non-normal or long-tailed.
 - Method: Use Mann–Whitney U test or Wilcoxon rank-sum test.
- Significance Level & Reporting Method
 - Significance level: $\alpha = 0.05$
 - Result Reporting:
 - Mean (or Median) $\pm$ Standard Deviation
 - Improvement Rate (%)
 - p-value
 - Effect Size + 95% Confidence Interval
- Interpretation Rules
 - If $p < 0.05$, the improvement rate meets the target ($\geq 15\%$), and the SLA ($\leq 90s$) is adhered to, the H2 hypothesis is judged as True.
 - If the increase in vehicle delay exceeds the guardrail ($> 5\%$), it is interpreted that while pedestrian wait times improved, the system requires calibration for practical use.

Success Criteria

- Mandatory Condition (Constraint): Maximum Pedestrian Wait Time ≤ 90 seconds.
- Performance Objective:
 - Pedestrian Average Wait Time: 15% reduction compared to baseline.
 - P95 Pedestrian Wait Time: Maintain or reduce.
- Side-effect Tolerance (Guardrail): Vehicle Average Delay increase rate $\leq 5\%$.

Hypothesis III

Improving Public Transit Travel Time

hypothesis Definition

If the signal control algorithm detects a bus object using Computer Vision(CV) and grants Transit Signal Priority(TSP) to the corresponding lane, the bus travel time will improve.

(Target: Improvement of $\geq 5\%$)

Metric Definition

- Bus:
- Bus Traversal Time & Route Segment Travel Time (Stop-to-Stop).
- Punctuality (Optional): Delay distribution compared to the scheduled baseline.
- General Vehicles:
- Utilize the Vehicle Average Delay metric defined in H1.

Verification Method

- Bus Travel Time: $\geq 5\%$ Improvement.
- General Vehicle Average Delay: $\leq 3\%$ Increase (or maintain near 0%).

Success Criteria & Objectives

- Objective: Improve bus travel time by 5% or more.
- Constraint: Maintain the increase rate of general vehicle average delay at 3% or less.

MVP Minimum Requirements (TSP Operational Rules)

- If a bus is approaching:
 - a. Green Extension: Extend the current green signal (Capped at Max +X seconds).
 - b. Early Green: Terminate the current phase early to switch to the bus phase (Provided that minimum red clearance and pedestrian safety intervals are satisfied).

MVP Development Plan

Objectives

- Implement algorithms that satisfy the hypotheses and configure the necessary Input/Output settings to run them on a Raspberry Pi 4, thereby building a prototype of the signal control system.
- Implement a signal controller that reduces the average intersection delay by:
 - Scheduling pedestrian phases using SLA to ensure the maximum pedestrian wait time does not exceed 90 seconds.
 - Dynamically weighting the vehicle green light duration based on real-time queue and inflow data verified through CV (Computer Vision).

Components

- SW: Setup of SUMO simulation for implementing and testing core algorithms.
- HW: Hardware setup for actual algorithm execution (running on Raspberry Pi 4) and I/O configuration.

Production Plan

- **Design a unified I/O structure to ensure that the algorithms for each hypothesis can operate on the Raspberry Pi 4.**
- **Configure MVP algorithms with the minimum features necessary to verify each hypothesis and test them within the simulation environment.**
- **Integrate the algorithms of validated hypotheses to evolve the system into a unified signal controller.**

H1 MVP Algorithm Design Plan

- Implement a dynamic signal algorithm that adjusts signal duration in proportion to real-time traffic. This involves identifying vehicle objects from image data using CV (Computer Vision) and SAM (Segment Anything Model).
- Algorithm Design

State Inputs

- Queue length per approach (number of vehicles)
- Cumulative wait time (or average wait time) per approach
- Current signal phase
- Elapsed time of the current phase
- Pedestrian request status
- Last pedestrian service time (time when the pedestrian signal was last active)

Signal Outputs (Actions)

- A0: Maintain current phase (within the maximum green limit)
- A1: Switch to the next phase (adhering to Green, Yellow, and Red rules)
- Adjust the action output duration in proportion to road traffic

Constraints

- Minimum Green Time: 20s
- Yellow Time: 3s
- All-Red Time: 1s
- Minimum Red Time: 120s

H2 MVP Algorithm Design Plan

- Implementation of an algorithm that schedules pedestrian phases to ensure the maximum pedestrian wait time does not exceed 90 seconds, while dynamically distributing vehicle green time based on real-time queue/inflow to reduce average pedestrian delay.
- Designed with future scalability in mind, such as reducing wait times in proportion to the number of waiting pedestrians.
- Basic Algorithm Design (Scalable)

Phase Settings:
Configure the entire signal system in units of Phases

- Vehicle Phases (2 Stages)
 - Phase V_NS: N-S Straight/Right Turn Green, E-W Red
 - Phase V_EW: E-W Straight/Right Turn Green, N-S Red
- Pedestrian Phases (2 Stages)
 - Phase P_NS: N-S Crossing WALK (+ FDW), Vehicles Red
 - Phase P_EW: E-W Crossing WALK (+ FDW), Vehicles Red
 - Insert Yellow / All-Red intervals between each phase.

Pedestrian Request Modeling

- Arrival of a pedestrian at the crossing start point (waiting area) is considered a ped_call event.
- Configure pedestrians in SUMO to wait in front of the crossing edge.

Signal Output

- Dynamically schedule waiting phases to ensure pedestrian crossing is enabled within 90 seconds upon a pedestrian request.
- Assign weights proportional to the number of waiting pedestrians.

H3 MVP Algorithm Design Plan

- Add an algorithm to the H1 MVP that grants Transit Signal Priority (TSP) to the corresponding road upon detecting a bus object in the image using SAM (in simulation, vehicle type set to 'bus').
- TSP Operation Algorithm Design

Trigger Conditions
(OR Logic)

- **Condition 1:** Bus enters the detection zone + The bus's direction is eligible for Green.
- **Condition 2 (Optional):** Intervene only when the bus is delayed beyond a certain threshold (e.g., delay > 30s).

Intervention Actions

- Action A: Green Extension**
 - If the signal for the bus's direction is currently Green, extend the duration up to a maximum of +10 seconds (Capped).
- Action B: Early Green**
 - If the signal for the bus's direction is currently Red, "advance to the next phase" only if the minimum Green time of the current phase has been satisfied.

Common Constraints

- No violation of pedestrian constraints (Maximum wait time must not exceed 90 seconds).
- "Consecutive intervention limit" to minimize negative impact on general traffic (e.g., Maximum of 2 consecutive interventions allowed).

Thank you!

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